Jinbin Hu | Curriculum Vitae

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I am currently a Post-doctoral from Department of Computer Science and Engineering, Hong Kong University of Science and Technology, Hong Kong SAR, advised by Prof. Kai Chen. Meanwhile, I am an Associate Professor in the School of Computer Science and Technology, Changsha University of Science and Technology, Hunan Province, Changsha, China. My current research centers on transport protocols and load balancing for modern datacenter networks, RDMA networking, learning-based network systems, communication optimization for distributed training, and programmable switching architectures*.

Education

- **PostDoc, Dept. of CSE, Hong Kong University of Science and Technology** — **Hong Kong SAR**
  Computer Science and Engineering — *1.2022–*
  Advisor: Prof. Kai Chen.
- **Ph.D., School of Computer Science & Engineering, Central South University** — **Changsha, China**
  Computer Science and Technology — *9.2016–12.2020*
  Advisor: Prof. Jiawei Huang.
- **M.S., School of Electronic & Information Engineering, Beijing Jiaotong University** — **Beijing, China**
  Microelectronics and Solid State Electronics — *9.2008–1.2011*
  Advisor: Prof. Xiaoguang Li.
- **B.S., School of Electronic & Information Engineering, Beijing Jiaotong University** — **Beijing, China**
  Electronic Science and Technology — *9.2004–7.2008*

Experiences

- **Changsha University of Science & Technology** — **Changsha, China**
  Associate Professor — *12.2021–*
  Teaching and researching in School of Computer and Communication Engineering.
- **Hunan Mechanical & Electrical Polytechnic** — **Changsha, China**
  Engineer — *8.2014–12.2021*
  Teaching and researching in School of Electrical Engineering.
- **National University of Defense Technology** — **Changsha, China**
  FPGA Verification Engineer — *8.2013–8.2014*
  Responsible for FPGA verification of multi-core CPU in Microelectronics Institute.
- **Empyrean Technology Co., Ltd** — **Beijing, China**
  IC Software Test Engineer — *1.2011–8.2013*
  Responsible for IC simulation software testing.

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Research Project

- **The National Natural Science Foundation of China**
  Study on Efficient and Scalable Transmission Control Mechanism for Lossy RDMA Networking in Data Centers, NO.62472050, 2025.1.1-2028.12.31. (PI)
- **The National Natural Science Foundation of China**
  Study on Transport Control in Data Center Lossless Network Based on Priority-based Flow Control, NO.62102046, 2022.1.1-2024.12.31. (PI)
- **The Young Scientists Fund A-class Project of Natural Science Foundation of Hunan Province**
  Research on Key Technologies for High Performance Transmission of Lossy RDMA Networking in Data Centers, NO.2025JJ20070, 2025.1.1-2027.12.31. (PI)
- **The Natural Science Foundation of Hunan Province**
  Study on Communication Optimization Technology for Distributed Deep Neural Network, NO.2022JJ30618, 2022.1.1-2024.12.31. (PI)
- **The Science and Technology Project of Hunan Provincial Department of Water Resources**
  Research on Digital Twin Technology for Three Dimensional Flood Evolution in Key Watershed Areas, NO.XSKJ2024064-36, 2024.9.30-2025.12.31. (PI)
- **The Scientific Research Fund of Hunan Provincial Education Department**
  Study on Load Balancing for Lossless Datacenter Networks, NO.22B0300, 2023.1.1-2025.12.31. (PI)

Representative Publications

1. **A High-Accuracy Sketch for Measuring Low-Entropy Flows in Distributed AI Training**
   *Jin Wang, Chenye Zhu, **Jinbin Hu***.* *6.2025*
   ACM ICPP, 2025. (CCF B)
2. **DMA-Sketch: A Fast and Accurate Sketch for Priority-Oriented Data Stream Processing**
   ***Jinbin Hu**, Houqiang Shen, Ying Liu, Jin Wang.* *6.2025*
   IEEE/ACM IWQoS, 2025. (CCF B)
3. **A Novel Load Balancing Mechanism Based on Deep Reinforcement Learning for Human-centric Applications in Datacenters**
   ***Jinbin Hu**, Wangqing Luo, Yi He, Shuying Rao, Fayez Alqahtani, Amr Tolba, Min Zhu, Jin Wang.* *6.2025*
   Human-centric Computing and Information Sciences (HCIS), volume 15, Article number: 35, 2025.
4. **FLB: Fine-grained Load Balancing for Lossless Datacenter Networks**
   ***Jinbin Hu**, Wenxue Li, Xiangzhou Liu, Junfeng Wang, Bowen Liu, Ping Yin, Jianxin Wang, Jiawei Huang, Kai Chen.* *5.2025*
   USENIX ATC, 2025. (CCF A)
5. **ARS: Adaptive Routing System for Heterogeneous Traffic in Industrial Data Centers**
   ***Jinbin Hu**, Rui Zhi, Jin Wang.* *5.2025*
   IEEE Transactions on Industrial Informatics (TII), 2025, DOI: 10.1109/TII.2025.3567376
6. **Proactive Transport with High Link Utilization Using Opportunistic Packets in Cloud Data Centers**
   ***Jinbin Hu**, Jiawei Huang, Zhaoyi Li, Yijun Li, Shuying Rao, Wenchao Jiang, Kai Chen, Jianxin Wang, Tian He.* *4.2025*
   IEEE Transactions on Mobile Computing (TMC), 2025. DOI: 10.1109/TMC.2025.3563182 (CCF A)
7. **SRCC: Sub-RTT Congestion Control for Lossless Datacenter Networks**
   ***Jinbin Hu**, Shuying Rao, Min Zhu, Jiawei Huang, Jianxin Wang, Jin Wang.* *12.2024*

IEEE Transactions on Industrial Informatics (TII), Volume 21, Issue 4, pp.2799-2808, 2024.

8. **Towards Fine-grained Load Balancing with Dynamical Flowlet Timeout in Datacenter Networks**
   Jinbin Hu, *Ruiqian Li, Ying Liu and Jin Wang.* *12.2024*
   Computer Networks, Volume 255, 2024.

9. **TA Novel Load Balancing Scheme based on PFC Prediction in Lossless Datacenter Networks**
   Jin Wang, Yi He, Wangqing Luo, Shuying Rao, ***Jinbin Hu****.* *10.2024*
   Human-centric Computing and Information Sciences (HCIS), Volume 14, 2024.

10. **Lightweight Automatic ECN Tuning Based on Deep Reinforcement Learning with Ultra-low Overhead in Datacenter Networks**
    Jinbin Hu, *Zikai Zhou, Jin Zhang.* *8.2024*
    IEEE Transactions on Network and Service Management (TNSM), 2024, 21(6): 6398-6408.

11. **AutoPipe: Automatic Configuration of Pipeline Parallelism on Shared GPU Cluster**
    Jinbin Hu, *Ying Liu, Hao Wang, Jin Wang.* *6.2024*
    In Proc. ACM ICPP, 2024. (CCF B)

12. **Enhancing Load Balancing with In-network Recirculation to Prevent Packet Reordering in Lossless Data Centers**
    Jinbin Hu, *Yi He, Wangqing Luo, Jiawei Huang, Jin Wang.* *5.2024*
    IEEE/ACM Transactions on Networking, 2024, 32(5): 4114-4127. (CCF A)

13. **RDMA Transports in Datacenter Networks: Survey**
    Jinbin Hu, *Houqiang Shen, Xuchong Liu, Jin Wang.* *5.2024*
    IEEE Network, 2024, 38(6): 380-387.

14. **TaLB: Tensor-aware Load Balancing for Distributed DNN Training Acceleration**
    Jinbin Hu, *Yi He, Wangqing Luo, Jiawei Huang, Jianxin Wang.* *4.2024*
    In Proc. IEEE/ACM IWQoS, 2024. (CCF B)

15. **Load Balancing with Multi-level Signals for Lossless Datacenter Networks**
    Jinbin Hu, *Chaoliang Zeng, Zilong Wang, Junxue Zhang, Kun Guo, Hong Xu, Jiawei Huang, Kai Chen.* *2.2024*
    IEEE/ACM Transactions on Networking, 2024, 32(3):2736-2748. (CCF A)

16. **A novel self-adaptive multi-strategy artificial bee colony algorithm for coverage optimization in wireless sensor networks**
    Jin Wang, Ying Liu, Shuying Rao, Xinyu Zhou, ***Jinbin Hu****,.* *11.2023*
    Ad Hoc Networks, Volume 150, 2023.

17. **Enabling Load Balancing for Lossless Datacenters**
    Jinbin Hu, *Chaoliang Zeng, Zilong Wang, Junxue Zhang, kun Guo, Hong Xu, Jiawei Huang, kai chen.* *10.2023*
    In Proc. IEEE ICNP, 2023. (CCF B, Best Paper Award)

18. **Adaptive Routing for Datacenter Networks Using Ant Colony Optimization**
    Jinbin Hu, *Man He, Shuying Rao, Yue Wang, Jing Wang and Shiming He.* *10.2023*
    In Proc. IEEE ICA3PP, 2023. (CCF C, Best Paper Award)

19. **RLB: Reordering-Robust Load Balancing in Lossless Datacenter Networks**
    Jinbin Hu, *Yi He, Jin Wang, Wangqing Luo, Jiawei Huang.* *8.2023*
    In Proc. ACM ICPP, 2023. (CCF B)

20. **Load balancing for heterogeneous traffic in datacenter networks**
*Shuying Rao, Ying Liu, Pradip Kumar Sharma, **Jinbin Hu***,. *8.2023*
Journal of Network and Computer Applications (JNCA), Volume 217, 2023.

21. **A Receiver-Driven Transport Protocol with High Link Utilization Using Anti-ECN Marking in Data Center Networks**
***Jinbin Hu**, Jiawei Huang, Zhaoyi Li, Jianxin Wang and Tian He.* *6.2023*
IEEE Transactions on Network and Service Management, 2023, 20(2):1898-1912. (CCF C)

22. **Load Balancing in PFC-Enabled Datacenter Networks**
***Jinbin Hu**, Chaoliang Zeng, Zilong Wang, Hong Xu, Jiawei Huang, Kai Chen.* *7.2022*
In Proc. ACM APNet, 2022. (CCF C)

23. **RPO: Receiver-driven Transport Protocol Using Opportunistic Transmission in Data Center**
***Jinbin Hu**, Jiawei Huang, Zhaoyi Li, Yijun Li, Wenchao Jiang, Kai Chen, Jianxin Wang and Tian He.* *11.2021*
In Proc. IEEE ICNP, 2021. (CCF B)

24. **Adjusting Switching Granularity of Load Balancing for Heterogeneous Datacenter Traffic**
***Jinbin Hu**, Jiawei Huang, Wenjun Lv, Weihe Li, Zhaoyi Li, Wenchao Jiang, Jianxin Wang and Tian He.* *6.2021*
IEEE/ACM Transactions on Networking, 2021, 29(5): 2367-2384. (CCF A)

25. **AMRT: Anti-ECN Marking to Improve Utilization of Receiver-driven Transmission in Data Center**
***Jinbin Hu**, Jiawei Huang, Zhaoyi Li, Jianxin Wang and Tian He.* *8.2020*
In Proc. ACM ICPP, 2020. (CCF B)

26. **CAPS: Coding-based Adaptive Packet Spraying to Reduce Flow Completion Time in Data Center**
***Jinbin Hu**, Jiawei Huang, Wenjun Lv, Yutao Zhou, Jianxin Wang and Tian He.* *10.2019*
IEEE/ACM Transactions on Networking, 2019, 27(6): 2338-2353. (CCF A)

27. **TLB: Traffic-aware Load Balancing with Adaptive Granularity in Data Center Networks**
***Jinbin Hu**, Jiawei Huang, Wenjun Lv, Weihe Li, Jianxin Wang and Tian He.* *8.2019*
In Proc. ACM ICPP, 2019. (CCF B, Best Paper Candidate)

28. **CAPS: Coding-based Adaptive Packet Spraying to Reduce Flow Completion Time in Data Center**
***Jinbin Hu**, Jiawei Huang, Wenjun Lv, Yutao Zhou, Jianxin Wang and Tian He.* *4.2018*
In Proc. IEEE INFOCOM, 2018. (CCF A)